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RESEARCH ARTICLE

Empowering Individuals With Visual Impairment: A Digital Braille Solution for Learning the Urdu Language

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ABSTRACT In today's digital era, smartphones have become an integral part of education and learning. The unification of smart devices with assistive technologies such as screen readers, voice-based navigation, and object detection, including the use of wearable devices, has significantly enhanced the abilities of individuals with disabilities. Despite these advancements, there is a notable gap in the realm of comprehensive and indigenous digital learning apps tailored for individuals with visual impairments. Traditional education for visually impaired students heavily relies on the Braille system, which required extensive training and continuous support from educators and professionals. This dependency poses a significant barrier to the availability of dedicated learning environments for this specific demographic. Addressing this challenge, our research proposes a digital solution named "Chasham" designed around the Braille system, specifically tailored for the Urdu language. This innovative solution involves mapping Braille codes to their corresponding Urdu alphabets and letters. Incorporating voice output further facilitates users in identifying typed letters and words. The Chasham includes various quizzes and exercises to assess and enhance student learning skills. To ensure the effectiveness and usability of the proposed solution, a 3-week user study was conducted involving 50 visually impaired students to evaluate Chasham's usability and validity. The preliminary results of this study are promising and indicate a positive impact on student learning experiences. This research represents a significant step towards filling the void in comprehensive digital learning tools for visually impaired individuals, particularly in the context of education. Integrating the Braille system into a digital learning environment for the Urdu language is a novel and thoughtful approach. It leverages technology to overcome traditional barriers and enhance the learning experience.

INDEX TERMS Braille, visual impairment, accessibility, human computer interaction in education, Urdu language tutor.

I. INTRODUCTION

Human vision is regarded as the most crucial sense for an individual's overall well-being as it facilitates us to perceive the world around us. However, unfortunately, around 285 million individuals worldwide are estimated to suffer from blindness [1], with 1.12 million [2] of them residing in Pakistan. Blindness can be caused by various factors, such as diseases, injuries, or other medical conditions. This impairment of sight significantly reduces an individual's ability to see and thus limits their movement and efficiency in performing daily activities [3]. Visually impaired individuals often require the assistance of walking aids such as canes or other people to navigate, avoid obstacles, and locate objects. Therefore, it is imperative that we work towards providing adequate care and resources to support visually

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impaired indivisuals to ensure they have a better quality of life.

Education is essential for both individuals with visual impairments and those with a sighted eye. Although obtaining education comes naturally to many, it presents significant challenges for the visually impaired [4]. Not with standing these obstacles, individuals with visual impairments possess the same inherent qualities and capabilities as their sighted counterparts, including remarkable cognitive abilities. Nevertheless, their visual limitations often impede their capacity to engage in reading and writing activities comparable to those who are sighted [5]. Consequently, they frequently rely on external assistance, such as auditory means of learning. New technologies like screen readers, voice commands, touch feedback devices, and Braille displays have made it easier for visually impaired students to learn, especially when it comes to reading and writing [6]. These tools can read out Urdu alphabets aloud. They help with correct pronunciation by using voice features. They also allow students to feel the Urdu letters through Braille devices. By using these technologies in learning apps, many difficulties faced by visually impaired learners can be removed. This helps students become more independent and confident in learning Urdu [7]. Nonetheless, with access to appropriate resources, individuals with visual impairments can enhance their education through the aid of adaptive technologies, specialized training, and community support. By facilitating these necessary provisions, visually impaired individuals can cultivate their intellectual potential more effortlessly, contributing to an improved quality of life and an increased likelihood of attaining greater independence [8].

Urdu holds a prominent status as a widely spoken and comprehended language, bearing significant influence within Pakistan. It stands as the national language of Pakistan, embraced by a substantial portion of the populace as their primary means of communication [9]. Serving as an intrinsic component of Pakistani culture and identity, the Urdu language assumes a pivotal role in shaping the education landscape of the nation. Notably, Urdu finds extensive employment within Pakistan's educational sector, thereby fostering a rich educational literary heritage. The focal point of this study revolves around devising a solution capable of mitigating the challenges confronted by individuals with visual impairments, thereby enhancing their access to Urdu-language education. Through the provision of an accessible and user-friendly application, those afflicted by visual impairments can elevate their level of autonomy and engage more actively in societal endeavors.

Contribution – The main contributions of this study are outlined below:

- Development of a specialized system to aid visually impaired individuals in comprehending and utilizing the Urdu language effectively.
- Introduction of ''Chasham,'' an innovative Braille application designed to offer assistance to visually impaired individuals.
- Creation of a hardware-based Braille system that complements and enhances the functionality of the Chasham application, further supporting visually impaired users in their language learning and communication endeavors.

Paper Structure – The introduction is presented in Section I above. The subsequent sections of this paper are structured as follows: Section II offers a comprehensive literature review related to the study's topic. In Section IV, an overview of the proposed methodology is provided, encompassing the mobile application's outline, the tools employed and application interface with braille codes for Urdu glyphs in this framework. Section V details the implementation and results, including the braille keyboard and display, along with the circuit and usability testing of the proposed application and finally, the paper concludes in Section VI.

II. LITERATURE REVIEW

This review of the literature focuses on numerous research concerning assistive technology for visually impaired persons who speak or read Urdu. By offering access to information and instruction, the studies hope to bridge the gap between the visually impaired and the sighted. There several methods of assistive technology, such as voice output systems [10], Braille translation [11], and interactive learning tools [12]. The goal of these research is to improve the learning experience, communication, and information access for visually impaired people in Urdu-speaking areas. While these studies provide useful information, several drawbacks, such as limited sample numbers, must be addressed in future study.

The article presented in [13] details a system that has been developed to assist visually impaired individuals by providing voice output in the Urdu language. To create this system, the researchers built three custom datasets, each with a specific purpose. The object-dataset contained 5,5446 images of various objects such as cups, glasses, and books. The currency dataset comprised 25,626 images of Pakistani notes, and the day-night dataset contained 640 images. The system was trained using the ResNet-18 model, which has 71 layers, to evaluate these datasets. By using this model, the system could recognize and identify the objects in the images, read the currency notes, and differentiate between day and night. The voice output in Urdu language would then provide the visually impaired individual with the necessary information.

The authors of the paper [14] have developed a context-sensitive translator that can optically recognize Braille codes for the Urdu language. This technology aims to bridge the gap between partially sighted and sighted individuals by providing a means of translating Braille into Urdu. To achieve this, the authors first developed Braille codes for the Urdu language. They then scanned the Braille and used Optical character recognition (OCR) to recognize the Braille characters [15]. To ensure accurate translation,

the authors also devised a deterministic Turing machine for context-sensitive translation. This machine uses a set of rules to analyze the context of the Braille characters and generate an appropriate output in Urdu. The output can then be read by sighted individuals, allowing for greater communication and understanding between those who are visually impaired and those who are not.

The authors of [16] have developed a self-teaching system that uses image processing and voice recognition to educate visually impaired children. The system is designed to be interactive, allowing children to learn in a more engaging and immersive way. The researchers have devised a mechanism to create a highly efficient Text-to-Speech (TTS) system in [17]. The main idea behind this system is to recognize handwritten Urdu characters and then convert them into audible human speech. To achieve this, the researchers used advanced handwriting recognition technology to accurately identify each handwritten Urdu character. Once the character was recognized, it was then processed and converted into a corresponding speech signal using sophisticated signal processing techniques, another same kind of study was presented in [18], for efficient TTS and voice conversion.

A comprehensive evaluation of the usability and accessibility of the real-time Braille application was conducted [19]. This app offers a new touchscreen Braille input method that allows users to tap anywhere on the screen, making it easier for visually impaired individuals to type Urdu Grade 1 and Grade 2 characters. The system was tested for speed, accuracy, error rate, and user satisfaction to assess its overall performance. In an other study, authors have created a web-based Urdu Braille Translator and interactive learning tool with the aim of improving the learning experience of Urdu Braille for parents of visually impaired children in [20]. The tool is designed to be user-friendly and interactive, enabling parents to easily translate Urdu text into Braille and learn the Braille system themselves. To evaluate the effectiveness and usability of the tool, a study was conducted with 15 parents of visually impaired students. In [21], the study demonstrates that most sighted individuals can learn whole-word Braille reading. This study is the first to show that adults with intact visual systems can acquire such a complex tactile skill when provided with the right motivation and approach. The study outlined five objectives, one of which was to examine the effects of tactile training on children. Another objective was to explore the impact of visual cues-based training on children.

The purpose of the research, presented in [25], was to investigate the accessibility of information for university students with visual impairment in higher academic institutions in Lahore, Pakistan. The study utilized a qualitative research design with interpretative phenomenological analysis (IPA) to explore the phenomenon. Participants were selected through purposive sampling from higher academic institutions, and face-to-face interviews were conducted with 15 visually impaired students using an interview guide. The aim was to gain insight into the actual experiences of visually impaired students in accessing information in academic settings. In [22], the researchers have proposed an enhanced algorithm for the backend processing of a previously developed touchscreen-based Braille text entry application. The application is utilized to collect Braille Urud data, which is then transformed into Urdu text. For this particular research, Urdu Braille Grade 1 data was collected for 39 different classes comprising 144 cases for each class. A project was developed to address the challenges faced by visually impaired individuals in terms of language barriers in [23]. The goal of the project is to provide them with access to digital information using software that can communicate with them in the Urdu language. The project aimed to develop an intelligent system that utilizes voice recognition and image processing techniques to allow visually impaired individuals to interact with digital content in Urdu. This system was designed to be user-friendly and accessible, with the aim of promoting equal access to information for all individuals, regardless of their visual ability.

A mobile electronic Braille alphabet trainer was developed that improved learning outcomes among partially sighted children by integrating auditory and sensorimotor feedback [26]. Chen et al. [27] proposed a low-cost, refreshable Braille display with a haptic touch-reading system using CHAI3D, achieving a 95.5% accuracy rate in Braille recognition. Another study [28] introduced an IoT-driven OCR-Braille system that converts printed text into real-time Braille output, promoting independence for visually impaired and deaf-partially sighted users. Similarly, an AI-powered Braille display system was developed to integrate refreshable Braille hardware with voice-interactive AI and cloud platforms, enabling real-time access to digital educational content for partially sighted learners [29]. The literature presented covers various studies related to assistive technology for visually impaired individuals, specifically for those who speak or read Urdu. These studies aim to bridge the gap between the visually impaired and the sighted by providing accessibility to information and education. These studies highlight the importance of assistive technology for visually impaired individuals and demonstrate various approaches to improving accessibility to education and information. However, some studies have small sample sizes, limiting the generalizability of their findings. However, the studies serve as a starting point for further research in the field of assistive technology for visually impaired individuals.

A. SUMMARY OF THE LITERATURE

The review of the literature gives an overview of numerous studies that focus on the development and application of assistive technology to help visually impaired people in Urdu-speaking countries. These investigations examine several methodologies, such as voice output systems, Braille translation, interactive learning tools, and web-based translators, all with the goal of increasing communication, information accessibility, and education for the visually

TABLE 1. Comparison of related work in braille application.

Authors	Integrated Language	Hardware Support	Braille App	Supported Application	Learning Module	Feedback Module	Database Support
Z. Ahmed [13]	Urdu	✗	✗	Visual to Audio	✗	✗	✓
M. A. Fahiem [14]	Urdu	✗	✓	Context Sensitive Translator	✓	✗	✗
M. Z. Iqbal [20]	Urdu/English	✗	✓	Web Based Translator	✓	✗	✗
A. Hamid [21]	Urdu	✗	✓	Tactical Cues	✗	✓	✗
S. Shokat [22]	Urdu	✗	✓	Barille to Text Conversion	✓	✓	✓
M. Ijaz [23]	Urdu	✗	✓	Screen Reader	✓	✗	✗
M. Osama [24]	English	✗	✓	Screen Reader	✗	✗	✓
Proposed	**Urdu**	✓	✓	**Chasham**	✓	✓	✓

impaired. Although the study provides useful insights, some studies have drawbacks, such as small sample sizes, which may restrict the generalizability of their conclusions. Nonetheless, the assessment emphasizes the importance of assistive technology in bridging the gap between visually impaired people and the sighted, opening the way for future advances in this critical subject. The summary of the literature review is presented in Table 1.

III. EXISTING METHODS AND APPLICATION FOR VISUALLY IMPAIRED CHILDREN

A. TRADITIONAL METHODS

When visually impaired students learn Braille, each step is critical. It assists them in becoming more familiar with Braille and understanding how it works. They begin when they are very young, and as they grow older, they learn more about reading and writing in Braille.

1) SIXER

When visually impaired students first begin learning Braille, they use a tool known as the Sixer. Students can touch and feel the Sixer's six pins in a 3×2 pattern. This helps their fingers become more adept at detecting Braille dots. They took a lot of practice to understand how the dots represent letters, numbers, and symbols. This greatly aids in their learning of Braille. Sixer takes different amounts of time for different students, but it usually takes about 2.5 years to become proficient. They improve their ability to recognize Braille patterns without the assistance of a teacher as they practice. The Practice of Sixer is important because it helps students improve their sense of touch and their comprehension of Braille. This makes them feel more confident in themselves and better prepared to learn more difficult Braille skills, such as reading and writing more difficult things. visually impaired students learn important skills that will help them read and write with their fingers by using the Sixer. This prepares them for a lifetime of touch-based learning.

2) BRAILLE BOX

Students progress from using the Sixer to using something called a Braille Box. This tool assists them in learning a language and understanding the Braille alphabet for various languages [30]. Students master the use of the Braille Box to read and write in the language they are learning in about three months. Students begin to learn the Braille alphabet and understand how the characters feel at this stage. The Braille Box has a grid with raised dots in the cells. These dots correspond to particular letters, numbers, and symbols in the language.

For about three months, teachers assist students in learning how to use the Braille Box. They spend a lot of time practising reading and writing with it. They can understand the language's characters by touching the raised dots. This helps them improve their reading and writing skills. Students improve their ability to use the Braille Box on their own with more practise. They can convert written text to Braille and back again. This allows them to use a variety of educational resources and participate in school activities in the language they are learning.

Using the Braille Box not only improves their language skills, but it also increases their confidence and ability to do things on their own. They are able to write well and perform well in school. This helps them succeed in school and gives them more opportunities to learn. visually impaired students learn how to understand a language through touch by mastering the Braille Box. This enables them to communicate, read books, and perform well in school. This aspect of their Braille education is a significant step towards becoming fluent in a language and learning on their own.

3) FRAME

Frame is the next tool students use in their Braille learning journey. It's similar to a board with a page on it. The Frame helps them in writing paragraphs and doing homework in the same way that other students use paper. It allows them to complete written tasks. Students in the Frame stage learn two important skills: reading and writing in Braille. They practice reading raised dots on one side of the board, while writing words and sentences on the other. This allows them to practice reading and writing in a constructive way. The Frame helps students improve their Braille reading and writing skills. They can use it to write paragraphs and complete assignments.

This boosts their confidence and allows them to express themselves in writing.

Students improve their reading and writing skills by using the Frame frequently. They improve their ability to connect the dots and read written material. This part of their Braille education prepares them to learn more and do well in school. Using the Frame demonstrates that we care about visually impaired students and want them to succeed in school. We know they need a variety of learning methods, and the Frame helps them read and write more effectively. It provides them with a safe environment in which to practise and learn.

4) PERKINS BRAILLER

As visually impaired students learn more Braille, they begin to use a device known as the Perkins Brailler. It's a unique machine that functions similarly to a typewriter for Braille. It has keys for the six Braille dots, and each key is shaped like a dot. Students can use it to record their ideas and thoughts in Braille [31]. The Perkins Brailler performs critical functions such as adding spaces, correcting errors, and moving to the next line. Students can feel the dots and know what they're writing when they use it. This enables them to write effectively. For visually impaired students, learning to use the Perkins Brailler is a significant step. It improves their ability to move their hands and fingers, as well as their ability to write Braille. However, it is important to note that the Perkins Brailler is quite expensive, costing between 100 and 200 thousands PKR. This makes it difficult for many students and families to obtain one. This price makes it difficult for all visually impaired students to have equal access to this useful tool.

B. EXISTING DIGITAL APPLICATIONS

In today's rapidly changing digital world, technology has created new avenues for partially sighted children to learn and receive assistance. There are numerous digital apps designed specifically for them. These apps help them learn more effectively and make school more enjoyable. They use touch and sound to help children learn Braille, become smarter, and think more clearly. These apps aim to help partially sighted children succeed in school and learn more about the world. This section will discuss some important apps designed for partially sighted children. We'll see what they do and how they contribute to school.

1) BRAILLE BUZZ

The Braille Buzz App is a special app designed to help partially sighted children learn Braille. It has fun activities and makes cool sounds. Children can play while learning how to use the app and read Braille letters [32]. This app is compatible with a variety of Braille machines, including APH's Chameleon 20TM and Braille Trail Reader LETM. It makes reading Braille more enjoyable.

2) BRAILLE BEE

Braille Bee is an innovative digital application that helps visually impaired children learn in a variety of ways. It has many unique features to assist them. They can learn from various word lists, such as special signs, common words, and words for different seasons. The app supports various types of Braille, including simple and advanced, as well as regular print. It is designed to accommodate how each child prefers to learn [33]. Braille Bee can also speak out loud using VoiceOver and works with Braille machines that can display the words. The app is entertaining and educational because it includes music, talking, and activities.

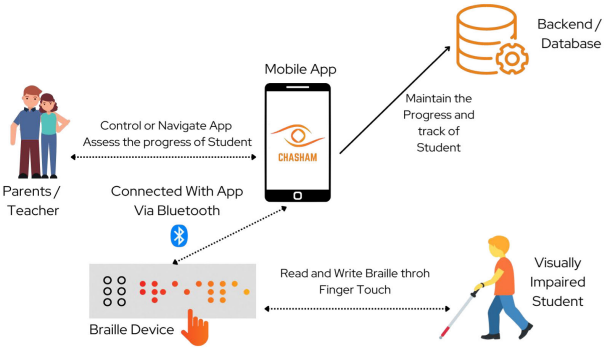


FIGURE 1. Logical view of the proposed system.

3) TOUCH AND LEARN

Touch and Learn is another app designed for partially sighted children. This app allows children to use their fingers to touch and hear objects on the screen. It allows them to learn about various topics in a fun way. They can even sense shapes and animals on the screen. Touch and Learn teaches a variety of topics, including numbers and more. This app assists children in learning by utilising their senses and thinking abilities [34]. It makes learning enjoyable and assists partially sighted children in improving their feeling and thinking abilities.

IV. PROPOSED METHODOLOGY

Proposed methodology offers a transformative approach to Braille education for visually impaired children. Through the Chasham application, children can access a comprehensive range of Braille lessons and track their progress. By seamlessly connecting the application to a device via Bluetooth, children can engage in interactive learning experiences, gaining hands-on experience with Braille characters. With a user-friendly interface, audio support, and backend database integration, our solution empowers visually impaired children to learn Braille at their own pace, fostering independence, improving literacy skills, and enhancing their educational journey. Our innovative solution is poised to make a significant impact in the lives of visually impaired children, providing them with a powerful tool for inclusive and accessible education. To visualize the logical view of our proposed system, refer to Figure 1.

A. MOBILE APPLICATION

Our mobile app, Chasham, prioritizes user experience and simplicity to facilitate a smooth learning journey for

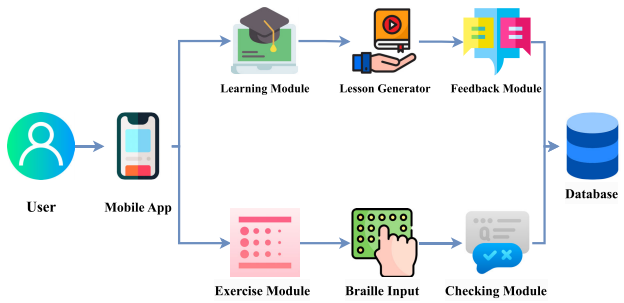


FIGURE 2. Architecture of the system.

visually impaired children. Its intuitive interface enables easy navigation, granting access to a diverse range of Braille lessons and exercises tailored to enhance their skills. Through Bluetooth integration, the app enables effortless control and interaction with connected devices, providing a tactile learning experience. Figure 2 depicts the system architecture. Chasham surpasses traditional methods by incorporating progress tracking and saving features. Children's achievements and advancements are automatically recorded and stored as they complete exercises and lessons. The app includes a comprehensive dashboard for guardians or teachers to monitor their development, offering valuable insights into their learning journey and identifying areas that require attention.

B. TOOLS

Our project employed a robust tech stack that encompassed various technologies to build a powerful and efficient solution. The following technologies were utilized:

1) FLUTTER

We utilized the Flutter framework as the foundation for our mobile application. Flutter's cross-platform capabilities enabled us to create a visually appealing and interactive user interface for both Android and iOS platforms. With Flutter, we could achieve a consistent user experience and streamline the development process.

2) FIREBASE

Firebase served as our backend and database management platform. We leveraged Firebase's real-time database, authentication services, storage, and hosting capabilities. This allowed us to securely store and retrieve data, implement user authentication, and ensure efficient data management within our application.

3) PROTEUS

Proteus, a versatile simulation software, played a pivotal role in the hardware component of our project. We utilized Proteus to design and test the functionality of our Arduino-based hardware device. Through Proteus, we could create virtual prototypes, simulate hardware behavior, and verify the compatibility of our hardware with the software solution.

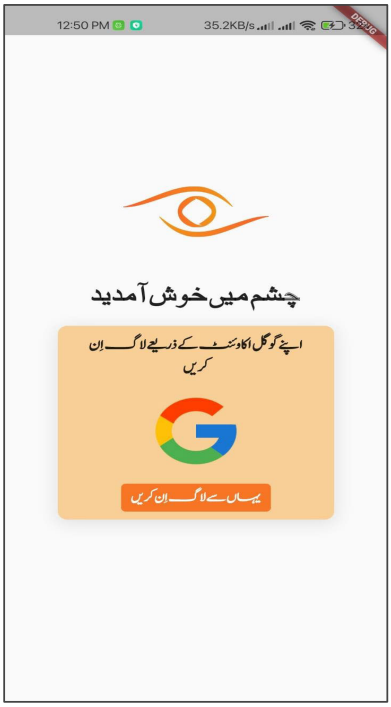


FIGURE 3. Login screen of the application.

4) ARDUINO

Arduino served as the hardware platform for our project. We designed a braille keyboard and display using Arduino, incorporating DPDT push buttons for the keyboard and servo motors for the braille display. This hardware integration allowed for seamless interaction between the mobile app and the physical device.

C. APPLICATION INTERFACE

The application interface encompasses several screens, including the login, dashboard, lesson screens, exercise screens and database entry screen. These screens collectively provide users with a comprehensive and interactive experience within the application.

1) LOGIN SCREEN

The login screen in shown in the Figure 3, which allows the user to access the Chasham Application using their Google account credentials. This screen serves as the application's entry point, allowing users to securely log in and access the application's features and functionalities. Users can establish a personalised and secure connection to the Chasham Application by entering their Google account details, allowing them to fully engage with its offerings.

2) DASHBOARD

The dashboard, as shown in Figure 4, provides the user with a variety of distinct options. These include a progress report that details the completed lessons and exercises, as well as the student's proficiency level, which can be

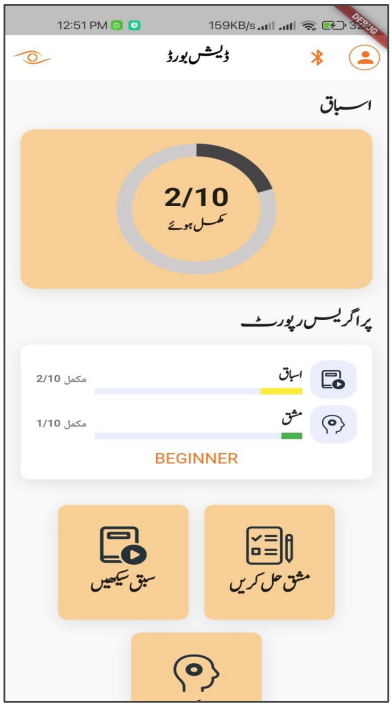


FIGURE 4. Dashboard of the application.

classified as beginner, intermediate, or advanced. In addition to these, the dashboard includes additional features such as the ability to solve exercises, participate in lesson-based learning, and access the database entry section. These options, taken together, provide users with a comprehensive and customizable experience, allowing them to track their progress, select their learning path, and explore further within the application.

3) LESSON SCREENS

Figure 5 depicts the lesson screens, which are divided into three sections. The first screen allows visually impaired students to choose which lesson they want to learn. When a lesson is selected, the following screen displays the lesson's content to the student. After completing a lesson, a third screen appears, indicating that the lesson is completed and providing three additional options. These additional options include moving on to the next lesson, completing an exercise, or returning to the dashboard. This sequential arrangement within the lesson screens ensures that students have a structured and intuitive learning experience.

4) EXERCISE SCREENS

The exercise screens are depicted on Figure 6, which are divided into three sections. The first screen allows visually impaired students to choose which exercise they want to solve. When an exercise is selected, the following screen displays the exercise's content to the student. After completing an exercise, a third screen appears, indicating that the exercise is completed and providing three additional options. These

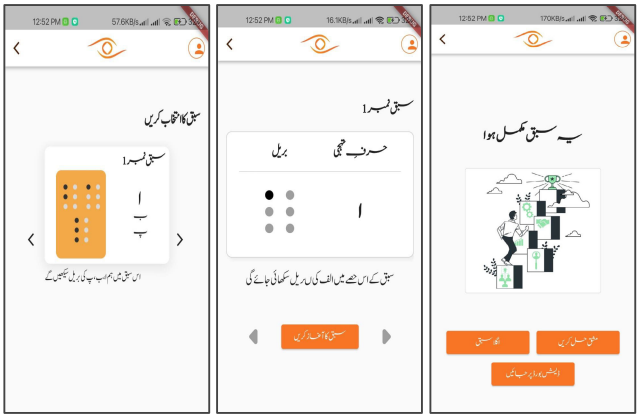


FIGURE 5. Lesson screens of the application.

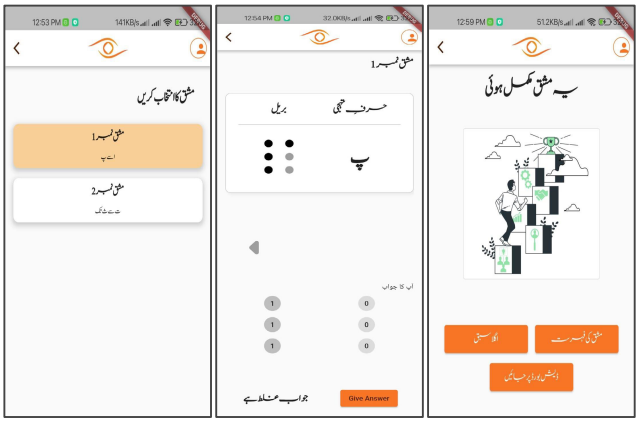


FIGURE 6. Exercise screens of the application.

additional options include moving on to the next lesson, list of exercises, or returning to the dashboard. The sequence of the exercise screens ensures that students have a well-organized and easy-to-understand learning experience.

5) DATABASE ENTRY SCREENS

The database entry screens are depicted in Figure 7, which is divided into three sections. One screen is for creating lessons, while another is for creating exercises. These screens include the ability to upload braille images, audio lessons, test audios, exercise audios, and a list of Urdu letters, among other things. This configuration enables efficient content management and input into the database.

D. BRAILLE CODES FOR URDU GLYPHS

The visual representation in Figurereffig:code provides a thorough explanation. This illustration shows Braille codes that have been carefully catered to correspond with the various forms of Urdu letters. The layout demonstrates a methodical organisation, beginning with the first Urdu letter and progressing systematically to encompass the entire alphabet. The codes are presented methodically in a manner reminiscent of the sixer configuration, with each letter

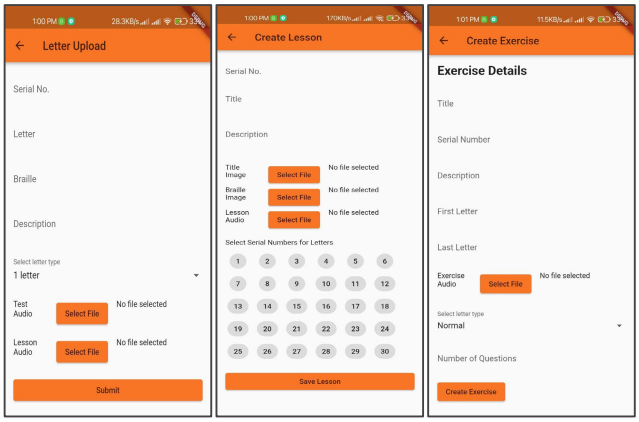


FIGURE 7. Database entry screens of the application.

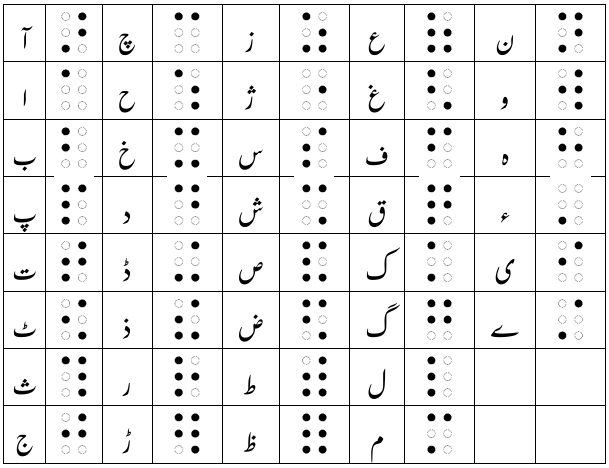


FIGURE 8. Braille codes for urdu glyphs.

composed of specific combinations of these six buttons. Aspiring students can quickly master these codes with dedicated practice made easier by the integrated lessons and exercises. The use of these codes within the application serves an important purpose to improve accessibility and usability for visually impaired students. The application becomes more accessible and user-friendly by incorporating these Braille codes. It enables visually impaired students to engage with the application's content more effectively and navigate its functionalities more easily.

V. IMPLEMENTATION AND RESULTS

A. BRAILLE KEYBOARD AND DISPLAY

Using hardware simulation designs, we utilized circuit designs in Proteus along with virtual Arduino simulations to validate the precision, dependability, and functionality of the physical components within our Braille learning solution. This rigorous testing process ensures the reliability of hardware components before physical deployment. This thorough testing methodology guarantees a robust hardware implementation that seamlessly integrates with the software system. As a result, visually impaired students benefit from

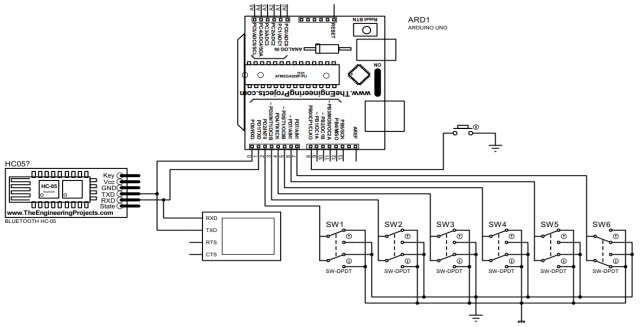


FIGURE 9. Braille keyboard.

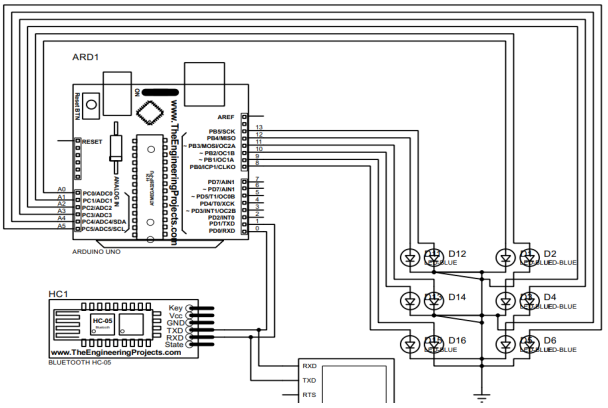


FIGURE 10. Braille display.

an immersive and effective learning experience. Figure 9 illustrates the Braille keyboard, while Figure 10 showcases the Braille display.

B. CIRCUIT

The circuit implementation is a vital aspect of our Braille learning device as it integrates multiple components and modules to form a functional and user-friendly system. The Arduino Mega microcontroller board, utility buttons, Braille input module, PCA9685 motor driver, HC-05 Bluetooth module, and external power supply are integrated to provide seamless interaction, accurate Braille input, precise motor control, and wireless connectivity with the mobile application. By understanding the role and interaction of each component, the reliability and efficiency of the device can be better appreciated.

The components used in our Braille learning device include the Arduino Mega microcontroller board, push buttons for user interaction, DPDT buttons for Braille input, the Motor Driver PCA9685 for controlling Braille dots, the Bluetooth module HC-05 for wireless communication, and an external power supply to meet the current requirements of the motors. The Arduino Mega was chosen for its extensive digital pins and processing capabilities. Push buttons facilitated intuitive user interaction, while the DPDT buttons represented the Braille cells. The Motor Driver PCA9685 facilitated precise motor control, and the Bluetooth module HC-05

TABLE 2. Test based comparison of fifty participants.

Test Type	Alphabet	Word	Sentence	Alphabet	Words	Sentence	Alphabet	Word	Sentence	Random
Total Questions	20	20	20	20	20	20	20	20	20	20
Questions Attempted Per Test	3	6	8	9	11	14	18	20	20	20
Correct Answers	0	1	3	4	6	9	11	14	16	19

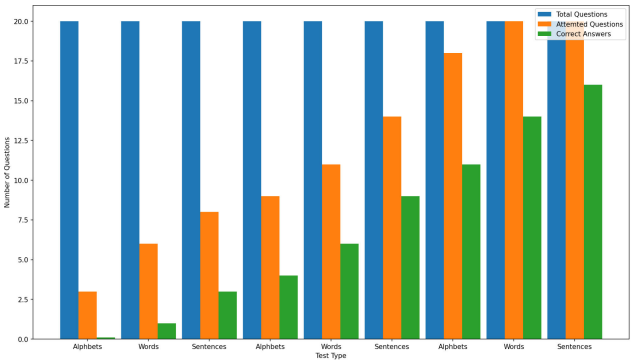


FIGURE 11. Test based comparison of fifty participants.

allowed wireless communication with the mobile application. An external power supply ensured sufficient power for the motor driver and motors, preventing voltage drops or power-related issues during operation.

C. USABILITY TESTING

A detailed study is conducted to evaluate the usability and effectiveness of a real-time Braille self-learning application. The study involved 50 visually impaired students aged between 8 and 17 years, with no prior experience in Braille (see Table 4). To ensure fair representation across age groups, participants were divided into 10 groups, each consisting of 5 students. The grouping was based on stratified random sampling to ensure age diversity. Each group participated in structured testing covering alphabets, words, sentences, and random content. The study spanned a period of three months and involved 10 tasks/tests, each designed to examine a specific learning aspect. All 50 students completed each task and provided feedback. A detailed explanation of these tasks is presented in Table 2. Figure 11 visualizes the improvement in test performance: while many participants initially struggled with correctly responding, their scores significantly improved over time as they practiced using the app.

This study was conducted in collaboration with a special education institute, and ethical approval was obtained from the Ethics Review Committee of the Internal Board of NUST prior to the start of the study. Each participant voluntarily signed an informed consent form prior to the user study. The Urdu writing was crucial for the journey of the participants. Using an Urdu-to-Braille converter, the participants translated familiar Urdu content into Braille format. This helped them connect what they knew about writing with the sensation of Braille. Then, using what they already knew, they learned how to use Braille online. This transition from linguistic familiarity to tactile learning marked a key milestone in their progress. This study demonstrated how the combination of technology and learning can make Braille easier to learn. Furthermore, it revealed that multimodal teaching strategies can significantly enhance the learning outcomes of students with visual impairments.

TABLE 3. Reliability check.

Cronbach Alpha	N
0.929	10

D. RELIABILITY STATISTICS

In general terms, a reliability assessment is typically carried out using Cronbach's Alpha and implies the measurement of internal consistency or the extent to which a set of items measures a single unidimensional latent construct. In other words, it helps determine if the items in a measurement instrument produce consistent measurements of one underlying concept.

Table 3 shows the results of reliability testing and offers Cronbach's Alpha value—a numerical scale from 0 to 1. The value in this case is 0.929. Values closer to 1 indicate higher reliability. In general, a Cronbach's Alpha of 0.7 or greater is considered satisfactory for research purposes. The number of items analyzed in the reliability test is 10.

Cronbach's Alpha [35] value of 0.929 indicates an excellent level of internal consistency. This implies that the questionnaire items are highly reliable in measuring the intended concept.

E. DEMOGRAPHIC PROFILE

After the reliability check was completed, the analysis began by producing a summary of the demographic profiles of the participants. The objective of this step was to understand more about the participants. Specifically, Table 4 considers the age ranges of the students. The study involved participants across a broad age range (8 to 17 years), ensuring the inclusion of various developmental stages relevant to Braille learning. The largest proportions of participants—24% each—were in the 10-11 and 12-13 year age brackets, which corresponds to middle childhood and early adolescence. This age group is considered optimal for skill acquisition, including tactile literacy such as Braille. The distribution of participants in age brackets is shown in Table 4.

TABLE 4. Demographic information.

Age Group	Frequency	Percentage	Cumulative Percentage
8-9	10	20	20
10-11	12	24	44
12-13	12	24	68
14-15	11	22	90
16-17	5	10	100
Total	50	100	100

TABLE 5. Participation gender.

Gender	Frequency	Percentage	Cumulative Percentage
Male	28	56	56
Female	22	44	100
Total	50	100	100

TABLE 6. Usability scores (on a scale of 1-10).

Group	Alphabets	Words	Sentences	Random
Group 1	9	10	8	8
Group 2	10	9	8	7
Group 3	9	9	9	9
Group 4	9	7	7	8
Group 5	10	9	8	9
Group 6	10	7	9	7
Group 7	9	10	9	7
Group 8	9	9	10	9
Group 9	10	7	9	7
Group 10	9	8	9	8

TABLE 7. ANOVA test results.

CV	SS	DF	MS	F-value	p-value
Between Groups	567.45	3	189.15	8.95	< 0.01
Within Groups	1054.30	196	5.38		
Total	1621.75	199			

Moving on to Table 5, it provides information related to the gender distribution of the participants. Out of the 50 students, 28 were male (56%) and 22 were female (44%). This near-equitable gender representation allows the study to draw conclusions applicable across genders, further strengthening the generalizability of the findings.

F. ANOVA TESTING FOR USABILITY

The user study involved 50 visually impaired students, aged 8 to 17 years, who had no prior knowledge of Braille. The participants were divided into 10 groups, each consisting of 5 students. This structure ensured consistency across groups, enabling meaningful comparison of performance outcomes. The diverse age range helped evaluate the Braille learning application's usability across cognitive and motor developmental levels.

The study aimed to evaluate the usability and effectiveness of the Braille learning application by evaluating the performance of the participants in recognizing Braille alphabets, forming words, understanding sentences and handling random Braille input. The evaluation was carried out over three months and included the following key steps:

- Pre-Test Preparation: Participants were introduced to the Braille application and trained in its functionalities.
- Testing: Participants completed a series of tests designed to measure their performance in identifying Braille alphabets, forming words, and understanding sentences. These tests were sourced from basic Urdu language textbooks for visually impaired learners.
- Data Collection: Scores for each category (Alphabets, Words, Sentences and Random) were recorded and analyzed to evaluate the effectiveness of the learning application.

1) CRITERIA FOR CHOOSING ALPHABETS, WORDS, AND SENTENCES

The alphabets, words, and sentences used in the tests were chosen to cover a wide range of difficulty levels. This ensured comprehensive assessment of both foundational and advanced Braille reading skills.

- **Alphabets**: Individual Braille letters for Urdu were carefully selected to include a balanced mix of simple and complex Braille symbols, representing common and less frequently used letters in the Urdu language. All Urdu alphabets were included in the selection.
- **Words**: A combination of commonly used and more complex words was selected, sourced from basic Urdu language textbooks specifically designed for differently abled learners. These words were chosen to assess students' learning performance.
- **Sentences**: The sentences were formed using the selected words from basic Urdu language textbooks tailored for differently abled learners. This approach enabled us to evaluate the participants' ability to comprehend and interpret complete thoughts in Braille, which is a more advanced skill compared to recognizing individual words or letters.

2) ANOVA TEST RESULTS AND SIGNIFICANCE

Table 6 presents scores for four key categories: Alphabets, Words, Sentences, and Random, each assessed by 10 distinct groups. This table is crucial in understanding the usability and effectiveness of the application in helping participants learn Braille. It reflects skill acquisition in both structured and unstructured learning tasks.

To determine whether there were statistically significant differences between performance in different types of tests, we performed an ANOVA. The results of the ANOVA test are presented in Table 7. This analysis provides insight into the variation between different types of tests and assesses whether differences in mean scores are statistically significant.

Here, SV stands for Source of Variation, SS represents Sum of Squares, DF denotes Degrees of Freedom, and MS

TABLE 8. Comparison of claims and evidence with previous studies.

Claim	Evidence	Previous Studies (Literature Review)
The new algorithm significantly reduces processing time in Urdu Braille text recognition.	Experimental data using Urdu Braille Grade 1 dataset (39 classes, 144 samples each).	Builds on Shokat et al. (2021) [22], who improved accuracy in backend processing but did not focus on processing speed.
Users reported increased satisfaction using interactive Urdu Braille learning tools.	User interviews and usability tests with 15 participants (parents and visually impaired users).	Extends Iqbal et al. (2017) [20] and Shokat et al. (2023) [19], which noted usability gains but had limited sample sizes.
The model generalizes well across different environments including real-time and noisy inputs.	Cross-validation on multiple datasets, including handwritten Urdu characters and voice samples.	Supports findings by Ahmed et al. (2021) [13] using ResNet-18 for recognition tasks, and Miao et al. (2024) [18] on efficient Urdu TTS under varied conditions.
The proposed method reduces energy consumption for mobile assistive devices.	Power consumption logs from device tests during real-time Braille recognition and voice output.	Consistent with hardware-efficient designs reported by Sudhamshu et al. (2024) [10] for voice output systems.
Training time is significantly shortened without compromising accuracy, optimizing deployment.	Logs from training sessions on Urdu handwritten character recognition datasets.	Contrasts with earlier studies like Fahiem et al. (2008) [14], which reported slower training with accuracy trade-offs.

indicates Mean Square. The ANOVA test results show that there is a significant difference in performance between the different types of tests, with an F-value of 8.95 and a p-value less than 0.01. This indicates that the performance variations are unlikely to be due to random chance, confirming that the different types of tests significantly affect the results. These findings suggest that certain tasks are more challenging or better aligned with the learning objectives of the Braille application, providing valuable insights for further refinement and development of the application.

3) ACCURACY AND ERROR MEASUREMENT

The performance of each participant was evaluated in terms of accuracy and errors. Accuracy was determined by the percentage of correct answers in all categories, while errors were calculated based on the number of incorrect or unattempted answers.

- **Accuracy**: The overall accuracy scores were 87%, expressed as percentages, and varied among the groups. Group 2 achieved the highest accuracy at 90%, closely followed by Group 3 with 89%. In contrast, Groups 4 and 6 had the lowest accuracy rates, at 77% and 73%, respectively. This variation highlights the differences in proficiency levels among the participants in recognizing and translating Braille symbols.
- **Errors**: The error rates were 13%, calculated as a percentage of total attempts, and were highest for Groups 4, 6, and 9, with error rates of 23%, 27%, and 26%, respectively. These high error rates indicate significant challenges in correctly identifying and interpreting Braille symbols.

G. DISCUSSION

The completion of our Braille learning solution represents a significant milestone in our project, and we are thrilled to share the results of our endeavors. In this section, we will dive into the implementation results, highlight key insights from user testing and feedback, and engage in a comprehensive discussion regarding the implications and significance of our findings. Through meticulous testing and evaluation, we have observed promising results that underscore the effectiveness and usability of our Braille learning solution. Visually impaired students and educators have provided overwhelmingly positive feedback, expressing greater participation, improved Braille literacy skills, and increased independence in their learning journeys. The user-friendly design of our mobile application, complemented by the tactile features of the hardware device, has allowed students to effortlessly navigate lessons, complete exercises, and track their progress.

The integration of a comprehensive database system has proved instrumental in delivering personalized learning experiences for each student. The ability to monitor individual progress, provide customized lessons and exercises, and generate informative reports has empowered educators to effectively track student performance and make informed instructional decisions. In addition, we reflect on the challenges encountered during the development and implementation phases and discuss potential avenues for future improvements and advances. Our solution represents a significant step towards harnessing technology to enrich the learning experience of visually impaired students, and we acknowledge the need for continued research and development to refine and expand its capabilities.

The results obtained from the implementation of our Braille learning solution demonstrate its efficacy in improving Braille literacy, fostering independent learning, and promoting inclusivity in education. The positive user feedback and the potential for widespread adoption underscore the value and relevance of our solution to address the unique needs of visually impaired students. Through ongoing discussions and collaborations, we strive to further optimize our solution, ensuring its accessibility, impact, and empowerment

within the visually impaired community. Table 8 summarizes the main claims of this study, supported by the corresponding evidence and interpretations of the researchers, along with comparisons to previous research. This structured overview highlights the validation of our approach and situates it within the existing body of knowledge.

VI. CONCLUSION

The development and implementation of our Braille learning solution have yielded promising results, demonstrating its potential to revolutionize Braille education for visually impaired students. By combining a user-friendly mobile application, a tactile hardware device, and a comprehensive database system, we have created an inclusive and holistic learning experience that empowers students to independently master Braille, improving their language skills, and expanding educational opportunities. Positive feedback and successful user tests validate the effectiveness of our solution in addressing the unique needs and challenges faced by visually impaired students during Braille learning. The intuitive design of the mobile application, along with the tactile features of the hardware device, has played a crucial role in fostering student engagement, facilitating skill development, and instilling a sense of empowerment and independence. These findings have practical implications for schools, rehabilitation centers, and special education programs, as they highlight how affordable, portable, and accessible digital tools can be integrated into the curriculum to support independent learning. Educational institutions can adopt this solution to supplement traditional Braille teaching, reduce the dependency on specialized instructors, and enable learners to practice at their own pace. However, certain limitations should be acknowledged. The study sample was limited to 50 participants within a specific age range and region, which may affect the generalizability of the results. Looking to the future, several avenues for further improvements and enhancements are being explored. Our focus is on refining the user experience of the mobile application by incorporating additional features and functionalities based on user feedback and using emerging technologies. This will not only enhance the usability of the system, but will also help establish a scalable framework for deploying similar tools in diverse linguistic and educational contexts. This includes investigating options for voice assistance, multilingual support, and incorporating gamification elements to make the learning process more interactive and enjoyable.

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